ENTERPRISE ACADEMIC GOVERNANCE SUITE

Edu Track – Student Academic Management System

A Unified, Role-Governed Full-Stack University Operating Environment with Real-Time Timetable Conflict Engines, Section-Scoped Coursework, and Automated Online Examination Proctoring

Technical Architecture & Executive Implementation Deck

Technology Stack:	Backend Engine:	Persistence Layer:
React 18 • TypeScript • Tailwind CSS • Vite	Node.js • Express • Socket.IO • JWT Auth	MongoDB Atlas + Dual-Store JSON Sync Engine

01. Executive Summary & Problem Landscape

Critical deficiencies in traditional higher-education academic administration software.

Legacy Academic ERP Challenges

- **Fragmented Portals:** Attendance, LMS, examination grading, and scheduling operate in disconnected silos, requiring duplicate manual data entry.
- **Timetable Scheduling Clashes:** Room, faculty, and cohort conflicts are caught retroactively rather than prevented at creation time.
- **Single Point of Failure:** Cloud database outages freeze campus administrative operations without a local fallback mechanism.
- **Attendance Fraud & Date Tampering:** Instructors backdating attendance or marking rosters outside scheduled operational windows.
- **Unregulated Online Examinations:** Lack of browser focus monitoring, lack of auto-evaluation, and insecure result release workflows.

The Edu-Track Modern Solution

- **Unified 3-Role Architecture:** Zero-friction role segregation across Institutional Administrators, Teaching Faculty, and Enrolled Students.
- **Autonomous Conflict Engine:** Proactive pre-commit validation of room, instructor, and student cohort overlaps for weekly timetables.
- **High-Availability Dual-Store Resilience:** Seamless automated fallback to local atomic JSON persistence if MongoDB Atlas drops offline.
- **Anti-Tamper Single-Day Attendance:** Hardware/server enforced restriction locking attendance exclusively to the current active date.
- **End-to-End Online Exam Pipeline:** Integrated proctoring (tab-switch detection), instant auto-grading, and faculty-governed result releases.

02. Full-Stack System Architecture & Dual-Store Engine

Multi-tier reactive architecture ensuring continuous availability and bi-directional synchronization.

CLIENT TIER (Frontend SPA)	GATEWAY TIER (API & Realtime)	PERSISTENCE TIER (Dual-Store)
• React 18 with TypeScript	• Node.js & Express REST Engine	• Primary: MongoDB Atlas Cloud
• Tailwind CSS + Lucide Icons	• JWT Bearer Authentication	• Fallback: Atomic JSON File Store
• Responsive Grid & Flex Layout	• Role Guard Middleware	• Auto-Switch on Network Failure
• Role-Governed Routing & State	• Socket.IO WebSocket Broadcasts	• Self-Healing Dual-Store Reconciler
• Recharts Academic Visualizer	• Multer Course Material Uploads	• Zero Client Operation Interruptions

High-Availability Dual-Store Resilience Mechanism

- **1. Detection:** Backend monitors MongoDB ping heartbeat. If unreachable (ECONNREFUSED / Network Timeout), state shifts to *Autonomous Local Store*.
- **2. Seamless Writes:** All CRUD operations route to structured JSON collections with ACID-like atomic rename file writes in `server/data/`.
- **3. Client Transparency:** REST endpoints return identical DTOs with `source: 'mongo' | 'json'`. The frontend displays live sync status pills without crashing.
- **4. Self-Healing:** Upon MongoDB reconnection, pending offline mutations synchronize back to cloud collections.

03. Role-Based Access Control (RBAC) Matrix

Strict enterprise permission boundaries isolating administrative, instructional, and student operations.

Capability / Operational Domain	Admin Portal	Faculty Portal	Student Portal
Manage Students, Faculty & Departments	Full Access (CRUD)	No Access (Hidden)	No Access (Hidden)
Weekly Timetable & Conflict Engine	Full Authoring & Schedule Publish	View Personal Faculty Schedule	View Enrolled Section Timetable
Semester Examination Timetable	Schedule & Publish All Depts	View Dept Exam Schedules	View Department Exam Timetable
Student Attendance Recording	View Institutional Analytics	Mark Session Attendance (Today Only)	View Personal Attendance %
Course Materials Distribution	System Storage Oversight	Upload Files (Section Scoped)	Download Section Materials
Online Exams & Proctoring	Audit Overall Assessments	Create Exams, Release Results	Attempt Once, View Results & Review
Official Grade Points & Transcripts	Publish Semester Grade Cards	Input Internal Assessment Marks	View Semester CGPA/SGPA Card

04. Class & Semester Exam Timetable Engine

Automated validation preventing cohort overlapping, room double-booking, and faculty collisions.

Weekly Class Timetable Architecture

- **Cohort Isolation:** Filtered dynamically by Department, Semester, and Section (e.g. CSE, Sem 6, Section A).
- **Persistent Matrix:** Stores periods across Monday through Saturday with Start/End times, Subject Code, Faculty ID, and Room.
- **Autonomous Conflict Detection:**
  - *Room Conflict:* Ensures no two classes occupy the same Room at the identical Day/Period.
  - *Faculty Clash:* Verifies the designated instructor is not assigned to another cohort simultaneously.
  - *Cohort Collision:* Guarantees student section cannot have simultaneous overlapping periods.
- **Immediate Broadcast:** Dispatches WebSocket notification `timetable_updated` to all connected student screens.

Semester Examination Timetable System

- **Institutional Scope:** University-wide schedule management for mid-term and end-semester examinations.
- **Department-Filtered Subject Dropdown:**
  - Eliminates manual code entry errors by reading registered departmental subjects directly from database.
  - Auto-populates official Course Title and Subject Code upon selection.
- **Universal Student Visibility:**
  - Once published by Admin, every enrolled student in that department immediately accesses the complete schedule.
  - Displays Examination Date, Time Window, Hall/Venue, and Official Conduct Instructions.
  - PDF export capability for student hall tickets and revision planning.

05. Faculty Section Isolation & Anti-Tamper Attendance

Enforcing assigned instructional boundaries and time-locked compliance in attendance tracking.

Faculty Section & Subject Scoping

- **Zero Cross-Section Leakage:** Instructors only see cohorts specifically allocated to them by the University Administrator.
- **Enrolled Students Directory:** Displays section-wise student rosters with register number, roll code, email, and live attendance %.
- **Assigned Subject Filtering:** Dropdown displays only subjects taught by the logged-in faculty, preventing unauthorized modifications.
- **Section Switcher Tabs:** Seamless navigation between assigned sections (e.g. CSE-A, CSE-B) with instant student counter badges.

Anti-Tamper Attendance Workflow

- **Today-Only Date Lock:** The attendance session date input is strictly locked to the current calendar date (disabling historical backdating and premature logging).
- **Streamlined Status Controls:** Binary **Present** and **Absent** toggles with 'Mark All Present' convenience accelerator.
- **Instant Batch Commit:** Saves session array with student IDs, attendance state, faculty signature, and server timestamp.
- **Automatic Aggregate Update:** Dynamically recalculates student total attended sessions, cumulative attendance %, and triggers low attendance warnings (< 75%).

06. Section-Scoped Coursework & Study Materials

Cloud document distribution and assignment submission tracking with granular section privacy.

Course Materials Upload & Distribution

- **File Upload Integration:** Instructors upload syllabus notes, lab sheets, and lecture slides via native file picker.
- **Section-Scoped Visibility:** Materials tagged to specific sections (e.g., CSE-B) appear exclusively in that section's student portal.
- **Persistent Metadata:** Stores document title, subject code, file size, MIME type, upload timestamp, and download URL.
- **Student Repository:** One-click secure asset retrieval with inline preview for PDFs and documents.

Assignments & Student Submission Ledger

- **Task Creation:** Faculty define assignment prompt, deadline, maximum marks, and submission format requirements.
- **Student Submission Vault:** Dedicated faculty review page showing exactly which students submitted, submission text, attached file links, and submission timestamps.
- **Grading & Evaluation:** Instructors award marks, enter feedback comments, and synchronize grades directly to student progress cards.
- **Real-time Sync:** Students receive instant alerts when coursework is assigned or graded.

07. Online Examination & Proctoring Pipeline

Secure, time-bounded multiple-choice examination system with real-time browser integrity proctoring.

Exam Authoring & Operational Lifecycle

- **Comprehensive Setup:** Title, Department, Semester, Section, Subject, Exam Date, Time Window, and Duration (mins).
- **Scoring Parameters:** Total marks, Passing cutoff, and configurable negative marking per question.
- **Dynamic MCQ Builder:** Instructors add questions, 4 option choices, designate correct key, and assign question weights.
- **State Machine Lifecycle:**
- • **UPCOMING:** Scheduled before start time.
- • **AVAILABLE:** Active exam window open for student attempts.
- • **EXPIRED:** Window closed; attempts barred.
- • **RESULTS RELEASED:** Scores and answer keys visible.

Student Exam Interface & Anti-Cheat Proctoring

- **Single-Attempt Enforcer:** Students who have already written the exam cannot re-attempt under any circumstance.
- **Live Countdown Timer:** Synchronized with server clock; auto-submits responses upon expiration.
- **Tab-Switch & Blur Monitoring:**
- • Detects when student unfocuses window or navigates to another browser tab.
- • Displays alert warning modal and increments proctoring infraction counter.
- • Logs total tab switches in final attempt payload for faculty integrity review.
- **Deterministic Auto-Evaluation:** Immediately compares student choices against answer keys on server submit.

08. Result Release Governance & Student Review

Faculty-controlled result publication guaranteeing transparent evaluation and perpetual access.

Faculty Result Release Workflow

- **Controlled Release Button:** Faculty review submission analytics, score distribution, and pass rate before hitting 'Release Results'.
- **Instant Status Transition:** Sets `resultsReleased = true` and shifts exam status to **RESULTS RELEASED (CLOSED)**.
- **Lockout of Future Attempts:** Automatically closes exam window to any unattempted students, safeguarding exam integrity.
- **Instant Broadcast:** Dispatches WebSocket notification triggering real-time UI refresh on all student dashboards.

Student Scorecard & Question Review

- **Unbroken Visibility:** Written exams remain permanently visible in student portal under 'Completed & Results' tab even after window expiration.
- **Detailed Scorecard:**
- • Marks Obtained vs. Total Marks (e.g. 20 / 20 M).
- • Percentage and Official **PASSED** / **NEEDS IMPROVEMENT** badge.
- • Submission timestamp and Recorded tab-switch count.
- **Interactive Question Breakdown:** Shows student choice, correct answer, and explanation for post-exam review.

09. Real-Time WebSocket Infrastructure & Event Bus

Socket.IO bi-directional communication ensuring instant collaboration across all academic stakeholders.

Socket Event Channel	Emitter Role	Target Audience	Operational Impact / UI Reaction
`timetable_updated`	Administrator	Dept Students & Faculty	Refreshes weekly timetable grid and room allocations without manual page reload.
`attendance_marked`	Teaching Faculty	Section Students	Updates live attendance counters, percentage gauge, and low attendance alerts.
`material_published`	Teaching Faculty	Cohort Section	Surfaces newly shared study PDF/link in Student Study Materials tab instantly.
`exam_results_released`	Teaching Faculty	Exam Students	Reveals final scores, pass/fail status, and question solutions to test takers.
`exam_scheduled`	Admin / Faculty	Target Students	Injects upcoming test card into student calendar and assessment notifications.

10. Security Architecture & Deployment Topology

Enterprise-grade security controls, input sanitization, and production cloud orchestration.

Application & Data Security Safeguards

- **Stateless JWT Bearer Auth:** Cryptographically signed tokens carrying user ID, role, department, and section claims.
- **RBAC Route Gateways:** Express middleware verifying roles before invoking controller logic (`verifyToken, checkRole`).
- **CORS & Payload Limits:** Whitelisted origins, parameterized queries preventing NoSQL injection, and sanitized file uploads.
- **Credential Protection:** Passwords hashed with BCrypt (10 salt rounds); secrets stored in strictly isolated environment variables.

Cloud Deployment & Continuous Delivery

- **Frontend CDN (Vercel):** React SPA deployed on global Edge Network with client-side routing rewrites and asset caching.
- **Backend Host:** Node.js + Express cluster on Linux server container with automated process management (PM2).
- **Database Hosting:** MongoDB Atlas M0/M10 replica set with TLS 1.3 encryption-in-transit and automated snapshots.
- **Repository Layout:** Monorepo structure separating `frontend/`, `backend/`, and `docs/` for streamlined maintenance.

11. Summary of Achievements & Future Roadmap

Transforming university academic operations through robust software engineering.

Key Project Achievements

- **100% Operational Cohesion:** Eliminated cross-system data synchronization gaps by uniting timetables, attendance, coursework, and exams.
- **Zero-Downtime Architecture:** Proven dual-store resilience guarantees 24/7 administrative continuity even during network partitioning.
- **Academic Anti-Fraud Guarantees:** Date-locked attendance sessions and tab-switch monitored examinations eliminate common malpractice vectors.
- **Exceptional User Experience:** Sub-100ms UI interactions, responsive modern design, and dark/light ambient theme support.

Future Expansion Roadmap

- **Mobile Applications:** Native iOS & Android portals built via React Native / Capacitor with push notifications.
- **AI Performance Diagnostics:** Predictive machine learning models identifying academically at-risk students for early intervention.
- **Biometric Attendance Hardware:** IoT fingerprint / facial recognition gateways syncing directly via REST API.
- **Automated University Accreditation:** One-click generation of NAAC / NBA compliance audits and outcome-based education (OBE) metrics.